



Foundational IT Skills

This laboratory course bridges basic computer use and professional engineering practice. It covers computer hardware, Linux file management, web communication, office automation, cloud collaboration, spreadsheet analysis, data visualisation and responsible use of generative AI tools.

REV2026

Semester I

Laboratory

Malayalam support

Printable handbook

Course code	1008
Semester	I
Course type	Laboratory
Contact hours	30
Credits	1
CIE / ESE	60 / 40
Modules	4

COURSE OVERVIEW

What this handbook covers

The structure follows the supplied official SITTTTR Kerala Revision 2026 syllabus. It is a learning handbook rather than a replacement for the official syllabus PDF. Use the module checklist to ensure that every prescribed point has been revised.

CO1

Identify functions and features of computers, operating systems and internet applications.

CO2

CO3

Construct reports and dashboards using spreadsheet formulas, PivotTables, slicers and charts.

CO4

Use generative AI and digital design platforms ethically for guided learning and visual communication.

Module roadmap

Module I

Computer fundamentals, Linux files and connectivity

Module II

Professional documents, spreadsheets, presentations and cloud

Module III

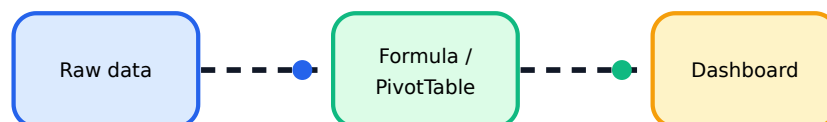
Data analysis, visualisation and AI slide generation

Module IV

GPT-guided learning, prompt engineering and graphic design

VISUAL CONCEPT

See the subject as a system



Clean → analyse → communicate → verify

MODULE I

Computer fundamentals, Linux files and connectivity

Computer system-ൽ Hardware എന്നത് physical parts ആണ്; Software instructions/programs ആണ്. Linux file system-ൽ folder structure ശരിയായി maintain ചെയ്യുന്നത് backup, search, version control എന്നിവ എളുപ്പമാക്കുന്നു.

Hardware exploration

Identify CPU, motherboard, RAM, HDD/SSD, power supply, display, keyboard, mouse and peripheral ports. Distinguish storage capacity, memory capacity and processor speed.

System and application software

Operating systems, drivers and utilities are system software; word processors, spreadsheets, browsers and CAD tools are application software. Explain open-source, proprietary and cloud software.

Linux GUI file management

Create, rename, copy, move, search and delete files/folders. Build a semester directory tree using subject codes and use meaningful filenames with dates or revision numbers.

Web search and connectivity

Use exact phrases, site:, filetype:, minus operator and date filters. Verify source authority, date and file type before downloading.

Email etiquette and LMS

Use clear subject lines, concise body text, proper attachments, CC for visibility and BCC to protect recipient privacy. Enrol and participate in Google Classroom or another LMS.

Rules / common mistakes

- Do not store important work only in Downloads.
- Use BCC for a large external recipient list.
- Verify URLs and file extensions before opening downloads.
- Maintain at least one backup copy.

Practice and self-learning

- Label the ports on a lab PC and compare PC/tablet specifications.
- Create the prescribed Backpack/Food/Tools folder exercise.
- Send a formal schedule-change email using To, CC and BCC correctly.

Professional documents, spreadsheets, presentations and cloud

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CO2

Office tools-ൽ manual formatting ഒഴിവാക്കി Styles, formulas, templates, slide master എന്നിവ ഉപയോഗിക്കണം. Cloud collaboration-ൽ version history ഉപയോഗിച്ചാൽ ആരാണ് എന്ത് മാറ്റിയത് എന്ന് കണ്ടെത്താനും പഴയ version restore ചെയ്യാനും കഴിയും.

Professional documentation

Create a formal report using page layout, styles, headings, table of contents, headers/footers, tables, images, captions and PDF export. Use styles rather than repeated manual formatting.

Spreadsheet structure and references

Understand rows, columns, worksheets, data types, relative reference A1, absolute reference \$A\$1 and mixed references. Validate data and avoid merged cells inside data tables.

Core functions

Use SUM, AVERAGE, COUNT, COUNTA, COUNTIF and IF. Check ranges, criteria quotes and whether text or numeric counting is intended.

Presentation design

Create a five-slide pitch deck with one message per slide, readable fonts, consistent alignment, limited animation and appropriate transitions. Use Slide Master for repeated identity information.

Cloud collaboration

Share with suitable permissions, co-edit in real time, use comments and suggestions, inspect version history and restore deleted content without overwriting collaborators.

Optional Overleaf introduction

Create a one-page document containing a title, list and mathematical formula. Understand source pane, compile process and PDF output.

Rules / common mistakes

- Keep raw data in a rectangular table with one header row.
- Never hard-code totals that should be formulas.
- Use fewer words and stronger visuals in slides.
- Do not grant edit access when view/comment access is sufficient.

Practice and self-learning

- Create an assignment cover sheet and professional biodata; export as PDF.

- Build the cricket spreadsheet using SUM, AVERAGE, COUNTIF and IF.
- Create a five-slide deck and configure Slide Master.
- Restore an accidentally deleted Google Sheet table using version history.

MODULE III

Data analysis, visualisation and AI slide generation

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PivotTable വലിയ dataset summary ചെയ്യാൻ ഉപയോഗിക്കുന്നു. Slicer interactive filter ആണ്. Chart തിരഞ്ഞെടുക്കുമ്പോൾ data relationship എന്താണ് എന്ന് ആദ്യം തീരുമാനിക്കണം; decorative chart കൊണ്ട് തെറ്റായ impression ഉണ്ടാക്കരുത്.

PivotTable workflow

Convert the source range to a clean table, insert PivotTable, place categorical fields in Rows/Columns and numeric fields in Values. Confirm SUM versus COUNT and refresh after source changes.

Slicers and interactive filtering

Insert a slicer for fields such as Region or Category. Explain how filtering changes visible totals while preserving source data.

Chart selection

Use column/bar for comparison, line for time trend, pie only for a small part-to-whole set, scatter for numeric relationship and combo charts only when scales are justified.

Chart quality

Add meaningful title, units and axis labels; avoid 3D distortion where exact comparison matters. Keep colour and legend use consistent.

Structured prompts for AI slides

Provide role, context, task, audience, constraints, slide count and required sources. Review every generated claim, image and citation before use.

AI-assisted research table

Ask an AI agent to collect structured information, but verify prices/specifications against primary sources and record retrieval date.

Rules / common mistakes

- Refresh PivotTables when source data changes.
- Do not use pie charts for many categories.
- AI output is a draft, not evidence.
- Remove personal or confidential data before uploading to AI tools.

Practice and self-learning

- Create a PivotTable and Region slicer from a 50-row sales dataset.
- Create pass/fail summary and suitable charts.
- Generate an eight-slide deck from a structured prompt and record generation time.

MODULE IV

GPT-guided learning, prompt engineering and graphic design

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CO4

Generative AI ഉപയോഗിക്കുമ്പോൾ prompt വ്യക്തമായിരിക്കണം; output verify ചെയ്യണം. Personal data, company drawing, password, confidential specification എന്നിവ public AI tool-ലേക്ക് upload ചെയ്യരുത്.

GPT-guided study assistant

Create a prompt that asks the model to teach one concept at a time, check understanding, give examples and stop for the learner's answer. Request uncertainty disclosure and source suggestions.

C-R-T-O framework

Context defines the situation, Role defines the assistant perspective, Task defines the required work and Output defines format, depth and constraints.

Technical interview practice

Ask five questions one by one, answer before the next question, request scoring against a rubric and ask for corrected model answers.

Ethical AI use

Avoid plagiarism, fabricated references, unsafe instructions, copyrighted asset misuse and disclosure of sensitive information. Cite AI assistance when institutional rules require it.

Graphic design platforms

Create a technical event poster or college festival logo with hierarchy, contrast, alignment, readable text and export settings appropriate for screen or print.

Open-ended experiment

Combine course skills into a new practical output such as an automated report, collaborative dashboard, AI-assisted study pack or event communication kit.

Rules / common mistakes

- Verify facts and calculations independently.

- Do not submit unedited AI output as original work.

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- Use licensed or self-created images.
- Keep prompt records for reproducibility.

Practice and self-learning

- Build a C-R-T-O prompt for a junior-engineer interview.
- Design a technical poster using an AI-enabled design tool.
- Plan and demonstrate one open-ended experiment mapped to a course outcome.

QUICK REVISION

Formula / rule bank

Rule / Formula 1

Relative reference A1; absolute reference \$A\$1

Rule / Formula 2

=SUM(B2:B21)

Rule / Formula 3

=AVERAGE(B2:B21)

Rule / Formula 4

=COUNT(B2:B21)

Rule / Formula 5

=COUNTIF(C2:C21,"Batsman")

Rule / Formula 6

=IF(B2>=50,"Selected","Waitlist")

Rule / Formula 7

PivotTable: Rows + Values + Filter/Slicer

College 1000

Rule / Formula 8

C-R-T-O: Context, Role, Task, Output

Rule / Formula 9

Email: To / CC / BCC

Rule / Formula 10

Version history: identify editor, inspect change, restore version

Worked checkpoints

Worked checkpoint 1

Use =COUNTIF(D2:D41,">=40") to count passes and =COUNTIF(D2:D41,"<40") to count failures.

Worked checkpoint 2

An absolute tax rate in H1 is used as =G2*\$H\$1 so the rate does not change when copied.

Worked checkpoint 3

For student marks by name, use a horizontal bar chart when names are long.

Worked checkpoint 4

Prompt: Context—Semester 1 IT lab; Role—patient tutor; Task—teach PivotTables with one dataset; Output—steps, check questions and a final exercise.

APPLICATION

Practical, drawing and self-learning work

No	Task	Required evidence
1	Hardware and software exploration.	Record procedure, observation/result and safety/quality checks.
2	Linux GUI file and folder management.	Record procedure, observation/result and safety/quality checks.
3	Advanced web search, email etiquette and LMS.	Record procedure, observation/result and safety/quality checks.
4	Professional report with styles and contents.	Record procedure, observation/result and safety/quality checks.
5	Spreadsheet formulas and references.	Record procedure, observation/result and safety/quality checks.
6	Five-slide presentation and Slide Master.	Record procedure, observation/result and safety/quality checks.
7	Cloud collaboration and version history.	Record procedure, observation/result and safety/quality checks.
8	PivotTable, slicer and data charts.	Record procedure, observation/result and safety/quality checks.
9	AI slide generation.	Record procedure, observation/result and safety/quality checks.
10	GPT study assistant prompt.	Record procedure, observation/result and safety/quality checks.
11	Technical poster/logo design.	Record procedure, observation/result and safety/quality checks.
12	Open-ended experiment.	Record procedure, observation/result and safety/quality checks.

EXAM PREPARATION

Expected questions and model-answer guides

These are syllabus-aligned practice questions, not a claim about the exact end-semester paper.

2 marks

Q1. Differentiate RAM, storage and processor.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q2. Classify system software and application software with examples.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q3. Write steps to create and manage a Linux directory tree.

Model-answer guide

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Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

2 marks

Q4. Explain advanced search operators with examples.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q5. Draft a formal email using CC and BCC correctly.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q6. Explain the purpose of styles and table of contents.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q7. Differentiate relative and absolute cell references.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q8. Write formulas using SUM, AVERAGE, COUNTIF and IF.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

5 marks

Q9. Explain how PivotTables and slicers work.

Model-answer guide

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Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q10. Select suitable charts for comparison, trend and part-to-whole data.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q11. Explain the C-R-T-O prompt framework.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

8-10 marks

Q12. List ethical and privacy precautions when using generative AI.

Model-answer guide

Answer using the definition/principle, a labelled diagram or formula where applicable, the ordered procedure/derivation, and one verification or application point. Cover the official module wording exactly.

COVERAGE AUDIT

Official syllabus coverage checklist

- Module I: Hardware exploration
- Module I: System and application software
- Module I: Linux GUI file management
- Module I: Web search and connectivity
- Module I: Email etiquette and LMS
- Module II: Professional documentation
- Module II: Spreadsheet structure and references
- Module II: Core functions
- Module II: Presentation design
- Module II: Cloud collaboration
- Module II: Optional Overleaf introduction
- Module III: PivotTable workflow
- Module III: Slicers and interactive filtering
- Module III: Chart selection

- **Module III: Chart quality**

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- **Module III: Structured prompts for AI slides**
- **Module III: AI-assisted research table**
- **Module IV: GPT-guided study assistant**
- **Module IV: C-R-T-O framework**
- **Module IV: Technical interview practice**
- **Module IV: Ethical AI use**
- **Module IV: Graphic design platforms**
- **Module IV: Open-ended experiment**

REFERENCES

Learning resources

1. Computer Fundamentals — P. K. Sinha, BPB Publications
2. IT Essentials: PC Hardware and Software — Cisco Networking Academy
3. Official learning tools referenced by the syllabus: Overleaf, Gamma, Manus and Canva

Primary authority for scope: supplied official SITTTTR Kerala Diploma Curriculum Revision 2026 syllabus PDF for Course 1008.